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Reduced graphene oxide–Nickel sulfide (NiS) composited on mechanical pencil lead as a versatile and cost-effective sensor for electrochemical measurements of bisphenol A and mercury (II)

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ABSTRACT

NiS is a highly promising transition metal sulfide versatile for a sensor material owing to its superior conductivity and stability, and herein we report its first incorporation into an electrochemical sensor. Because exposure to electrolyte under electrochemical impact can easily deform NiS, thereby decreasing its electroactivity and measurement reproducibility, a strategy for optimally integrating NiS into sensors is critically necessary. For this purpose, NiS was initially firmly affixed to a mechanical pencil lead (MPL) by means of hydrothermal deposition (with the result designated as MPL-NiS). MPL, a commercially available carbon-based material with consistent quality, was adopted to make the sensor cost-effective and easily prepared. Then, to prevent direct exposure of NiS to samples during electrochemical measurement and to make the sensor surface more reactive for wide variety of analytes, reduced graphene oxide (rGO) was electrodeposited on the NiS surface to construct a final structure of MPL-NiS/rGO. In summary, NiS, a *p*-type semiconductor with positive charge, was effectively composited and sandwiched with negatively charged rGO and MPL by means of mutual electrostatic interaction. When the developed MPL-NiS/rGO sensor was used to separately measure bisphenol A (BPA) and mercury- Hg^{2+}, the sensitivity and sensor-to-sensor reproducibility were comparable with or superior to those of previously reported sensors.

1. Introduction

Recently, various transition metal sulfides (M_xS_y) have been developed as versatile materials for uses in such as solar cells [1,2], supercapacitors [3], photon detectors [4], lithium ion batteries [5], transistors [6], light-emitting diodes [7], water splitting media [8], and fuel cell catalysts [9]. Among them, nickel sulfide (NiS) has drawn attention for its remarkable electronic properties such as its superior conductivity and its electroactivity comparable with that of Pt [10]. In addition, a recently reported simple solution-based synthetic method for NiS deposition has made the use of NiS more competitive.

As mentioned above, NiS is an inexpensive material that shows strong promise for electrochemical sensing owing to its high conductivity and chemical stability. However, research on this use has been scant thus far and exploration of NiS as an electrochemical sensing material is at early stage. To incorporate NiS into sensing systems, it is first necessary to resolve one critical issue. Namely, direct use of NiS as a sensor element is impractical because the structure of NiS is unstable during electrochemical scanning [11,12] due to slow diffusion of Ni^{2+} out of the lattice, which leaves behind sulfur-rich NiS that can degrade further ($NiS \rightarrow NiS_2 + Ni^{2+}$ and $NiS_2 \rightarrow S_2O_3 + Ni^{2+}$). This degradation decreases the electroactivity, narrowing the potential window and worsening the measurement reproducibility. An effective strategy to resolve this problem is essential for adopting NiS in electrochemical sensing.

Herein we demonstrate a NiS-based electrochemical sensor for the first time. In the sensor, NiS was integrated with mechanical pencil lead (MPL) and reduced graphene oxide (rGO). NiS is known to be a positively charged *p*-type semiconductor, so negatively-charged materials able to enhance electrochemical sensing performance can be easily integrated with it by means of electrostatic interaction. Initially, to fabricate a cost-effective sensor, MPL with weak negative surface charges [13,14] was simply used as a substrate upon which NiS was easily deposited (to form MPL-NiS) by means of a hydrothermal method

Abbreviations: MPL, mechanical pencil lead; rGO, reduced graphene oxide; BPA, bisphenol A

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[10]. MPL was used because it is a cheap, commercially available carbon-based material that has consistent quality and thus has shown high versatility in various electrochemical measurements. With this merit, we previously demonstrated a mechanical pencil lead-supported carbon nanotube/Au nanodendrite sensor for As (III) detection [15]. Next, to avoid direct exposure of the deposited NiS during electrochemical analysis to surrounding electrolyte as well as to make the sensor more reactive in measurements of a wide variety of analytes, reduced graphene oxide (rGO) was electrodeposited on the NiS (to form MPL-NiS/rGO). rGO is known to form ultrathin and durable conducting layers suitable for diverse electrochemical measurements [16–18]. By means of the above procedure, NiS was firmly held between negatively charged rGO and MPL layers, so we expected the resulting MPL-NiS/rGO sensor to be sensitive, rugged, and cost-effective. The structure of the MPL-NiS/rGO sensor was characterized using relevant methods and the electrochemical behaviors of MPL, MPL-NiS, and MPL-NiS/rGO were examined together.

We evaluated the utility of the MPL-NiS/rGO sensor for measuring two separate and well-known analytes, namely bisphenol A (BPA) and Hg^{2+}; BPA is an endocrine disruptor [19] and Hg^{2+} is a harmful heavy metal ion [20]. For the detection of BPA, prepared samples in the concentration range from 0.1 to 100 ppb were initially analyzed using anodic stripping voltammetry (ASV). Then, real BPA samples extracted from thermal receipt paper and a digital video disk (DVD) were analyzed, and their determined BPA concentrations were compared with those determined by means of HPLC. The detection performances thus determined were compared with those reported in previous publications regarding the use of various types of electrochemical sensors. Next, the sensor was further used to measure Hg^{2+} samples in a low concentration range from 5 to 100 ppb, again using ASV, and the resulting analytical performances were evaluated.

2. Material and methods

2.1. Preparation of MPL-NiS/rGO sensor

Cylindrical MPLs (diameter: 0.2 mm; length: 60 mm) were purchased at a local stationery store. One was cut to 20 mm length and placed vertically inside a tube containing $Ni(NO_3)_2$ (0.1 M) and thioacetamide (0.1 M) in ethanol. The tube was placed in a temperature bath held at 80 °C to allow the formation of NiS on the MPL (MPL-NiS). This procedure was replicated with the use of various hydrothermal reaction times ranging from 7 to 12 h [10]. Next, a constructed MPL-NiS was immersed into a dispersion of rGO in PBS electrolyte (1 mg/mL) and subjected to −1.0 V electrical potential for 180 s using a potentiostat (eDAQ, Australia) allowing rGO electrodeposited on the NiS surface to finally obtain a MPL-NiS/rGO sensor. The constructed MPL-NiS and MPL-NiS/rGO structures were examined by SEM (Hitachi S-4800 SEM, Japan).

2.2. Measurement of BPA and Hg^{2+} using MPL-NiS/rGO sensor

BPA powder (99.9%, Sigma-Aldrich) was dissolved in ultrapure water to make a 100 ppm stock solution. BPA samples with concentration range from 0.1 to 100 ppb were prepared using the stock solution immediately before each measurement. Each sample was mixed with 40 mL PBS 1 M (supporting electrolyte) and the pH was subsequently adjusted from 7 to 10 by adding volume of 1 M NaOH [21]. For real sample pretreatments, a DVD and thermal paper were each cut into 5 mm × 5 mm pieces. Then, 5.0 g of the DVD and 0.1 g of the paper were separately placed into 50 mL beakers and extracted by adding 10 mL of ultrapure water. Extraction was carried out under sonication for 30 min in a sonication bath with room temperature. The resulting solution was then carefully filtered by 0.45 μm filter membrane. The extracted solutions from the DVD and the receipt paper was diluted 200 and 10,000 times, respectively, for the measurement. The dilution was to make the concentration of both samples within a calibration curve range. For Hg^{2+} measurement, samples of various Hg^{2+} concentrations (from 5 to 100 ppb) were prepared from a standard Hg^{2+} solution (1000 ppm, analytical grade, purchased from Merck). A mixture of 0.1 M KCl and HCl at pH 3.0 was used as a supporting electrolyte for Hg^{2+} analysis [22]. Diluted Hg^{2+} samples of desired concentrations were prepared immediately prior to their measurement.

Both BPA and Hg^{2+} samples were analyzed by means of ASV. In BPA analysis [21], sample enrichment was previously performed by applying −0.2 V (vs. Ag/AgCl) to the sensor with stirring over 240 s, and voltammograms were acquired with scanning over the voltage range from 0 to +1.0 V. In the analysis of Hg^{2+} samples, −0.2 V was applied to a sample with stirring over 180 s and, a voltammogram was acquired over the range from −0.2 to +1.0 V. All electrochemical measurements were carried out at room temperature using Ag/AgCl and Pt wire as the reference and counter electrodes, respectively.

The employed HPLC (Jasco MD-2010) was equipped with CWN column (Athena C18-WP, reversed phase), a multi-wavelength UV/Vis detector and a quaternary gradient pump. Absorbance at 280 nm was used to construct a chromatogram. The mobile phase was composed of acetonitrile, de-ionized water and acetic acid with the volume ratio of 60:39:1. The flow rate was 1.0 mL min^{-1} and the same injection volume was 20 μL. Standard BPA solutions in a concentration range from 0.05 ppm to 220 ppm were prepared and used to build a calibration curve.

3. Results and discussion

3.1. Examination of MPL-NiS/rGO structure and its electrochemical behavior

SEM images were obtained from MPL-NiS (NiS hydrothermally deposited on MPL) samples prepared with six different durations of deposition from 7 to 12 h (Fig. 1). Also, SEM image of MPL is shown for comparison. NiS was sparsely deposited on the surface after the 7 h reaction, and the coverage increased with longer reaction time. After 11 h of deposition, the surface was well covered with diverse NiS nanostructures such as particles and clusters (Fig. 1f), and after 12 h, the distinct nanostructures had disappeared owing to subsequent agglomeration (Fig. 1g). To additionally confirm the formation of NiS, the 11 h sample was subjected to EDS elemental analysis, which indicated that the atomic ratio between Ni and S was nearly equal, clearly indicating the formation of NiS without noticeable impurities (refer to Fig. S1 in the Supplementary data). Then, the diffusional electrode areas (A_{diff}) of MPL-NiS structures constructed using various durations of hydrothermal deposition were calculated using the Randles–Sevcik equation [23], expressed as follows.

$$i_p = 2.69 \times 10^5 n^{3/2} D^{1/2} A_{diff} C v^{1/2}$$

Here, i_p is the peak current corresponding to the reduction of redox species, n is the number of electrons transferred in the redox event, D is the diffusion coefficient of the analyte, C is the molecular concentration of the analyte, and v is the scan rate in V s^{-1}. To determine A_{diff} values, cyclic voltammograms of the redox couple of $[Fe(CN)_6]^{3-}/[Fe(CN)_6]^{4-}$ were acquired using the prepared MPL-NiS structures (shown in Fig. S2a). The C and D values used for $K_3[Fe(CN)_6]^{4-}$ were 5×10^{-3} M and 7.5×10^{-6} cm^2 s^{-1}, respectively. The A_{diff} values of the structures prepared with the NiS deposition times of 7, 8, 9, 10, 11, and 12 h were 1.37, 1.52, 1.62, 1.73, 1.91, and 1.66 cm^2, respectively. This result again confirmed that the MPL-NiS prepared using the 11 h deposition duration was the most optimal as also expectable by examining the SEM images (Fig. 1) and the corresponding 11 h MPL-NiS structure was used for subsequent sensor preparation step. Next, to complete the construction of the MPL-NiS/rGO sensor, rGO was simply electrodeposited on the NiS surface of MPL-NiS with the help of mutual electrostatic

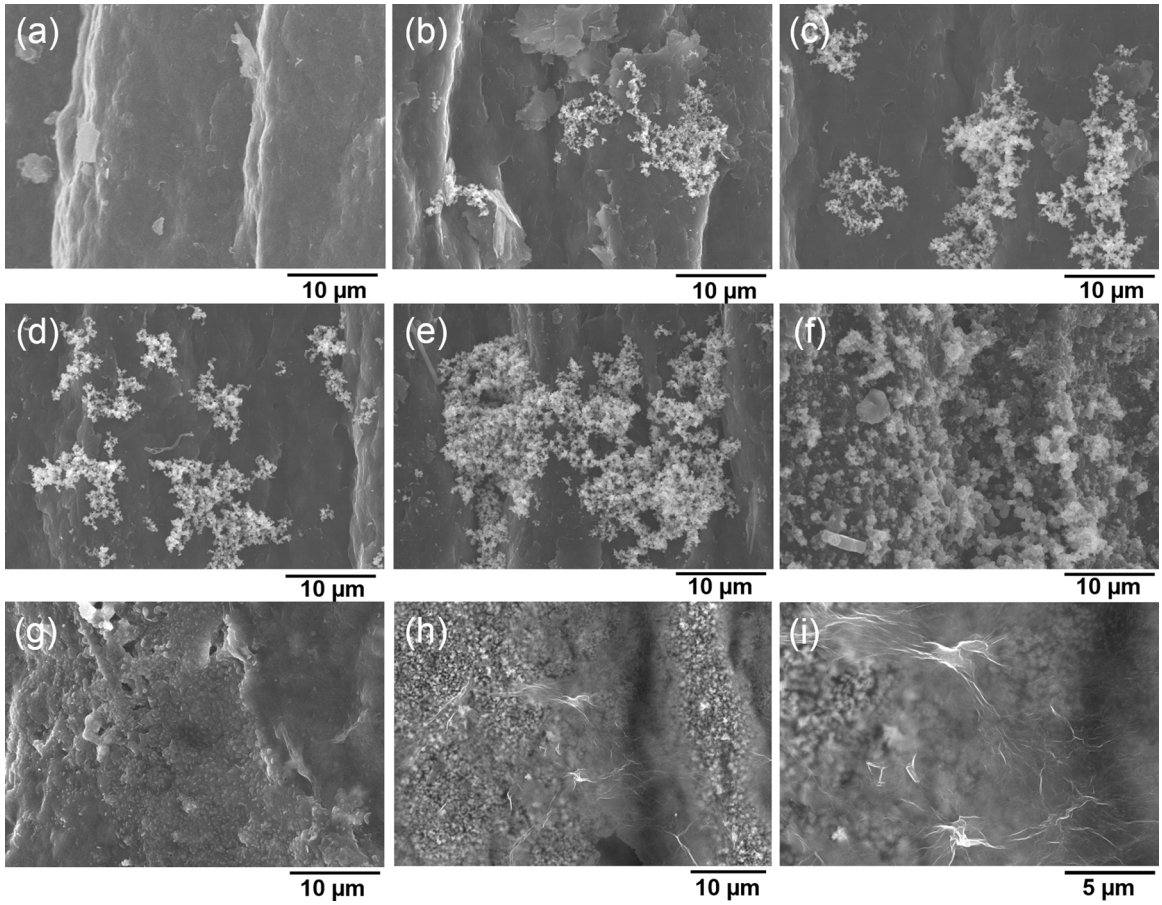

Fig. 1. SEM images of (a) MPL and MPL-NiS structures prepared with various durations of NiS deposition: (b) 7, (c) 8, (d) 9, (e) 10, (f) 11, and (g) 12 h. Also, SEM images MPL-NiS/rGO structures ((h) and (i)) are also shown.

interaction by applying the potential of −1.0 V. SEM images clearly showed the sheet-like cover of rGO on NiS with some wrinkles. (Fig. 1h and i). For reference, cyclic voltammograms of the same sample acquired again using MPL-NiS/rGO structures prepared with different NiS deposition (7–12 hours) are also shown in Fig. S2b. The MPL-NiS/rGO prepared with the NiS deposition time of 11 h showed again the highest peak current.

Fig. 2 shows cyclic voltammograms (CVs) of the 5×10^{-3} M $[Fe(CN)_6]^{3-}/[Fe(CN)_6]^{4-}$ redox reaction acquired using the MPL, 11 h MPL-NiS, and 11 h MPL-NiS/rGO structures. The peak intensity observed using the MPL-NiS/rGO was respectively 3.2 and 1.6 times greater than those observed using the MPL and MPL-NiS. This observed increase in current intensity arise from the sequential incorporation of rGO, a more conducting material, to NiS. Also, using the MPL-NiS, and MPL-NiS/rGO structures, CVs of the same sample was consecutively measured with 20 cycles and the variations of peak intensity were examined, as shown Fig. S3. After 20 cycles, the intensity was decreased by 21.9 and 2.6% in the case of using MPL-NiS and MPL-NiS/rGO electrodes, respectively. This result clearly confirms that NiS/rGO is much more stable than NiS itself during electrochemical analysis.

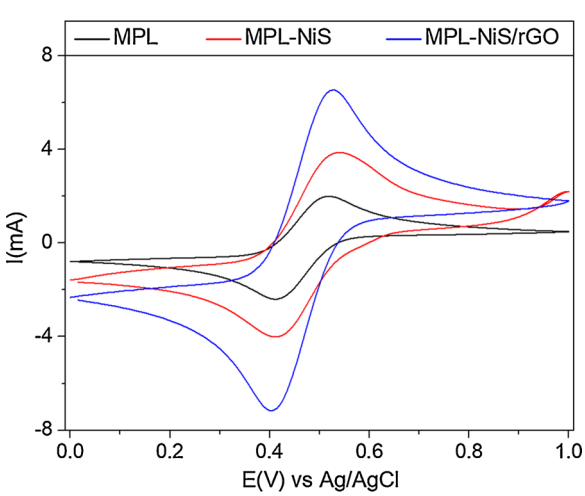

Fig. 2. Cyclic voltammograms of 5×10^{-3} M $[Fe(CN)_6]^{3-}/[Fe(CN)_6]^{4-}$ redox reaction acquired using the MPL, MPL-NiS, and MPL-NiS/rGO structures.

3.2. Measurement of BPA samples using MPL-NiS/rGO sensor

Fig. 3(a) shows cyclic voltammograms of a 0.1 ppm BPA sample separately measured using MPL, MPL-NiS and MPL-NiS/rGO sensors without any enrichment step. The oxidation peak of BPA, located around 0.6 V, was apparent when the MPL-NiS/rGO sensor was used, but not when the other two structures were used. This means that the NiS surface did not strongly attract the BPA; instead, the observed peak was attributed to the interaction between BPA and the π-conjugated electronic structure of rGO [21]. This is another advantage of incorporating rGO to NiS, which enables to attract diverse analytes especially possessing aromatic rings to the sensor surface for electrochemical reaction.

Next, stripping voltammograms of 14 BPA samples (concentration range 0.1–100 ppb) were acquired by ASV method using the MPL-NiS/rGO sensor (Fig. S4). The top inset of Fig. 3 shows the correlation between BPA concentration and peak intensity in the tested concentration range; the bottom inset shows the variation of peak intensity below

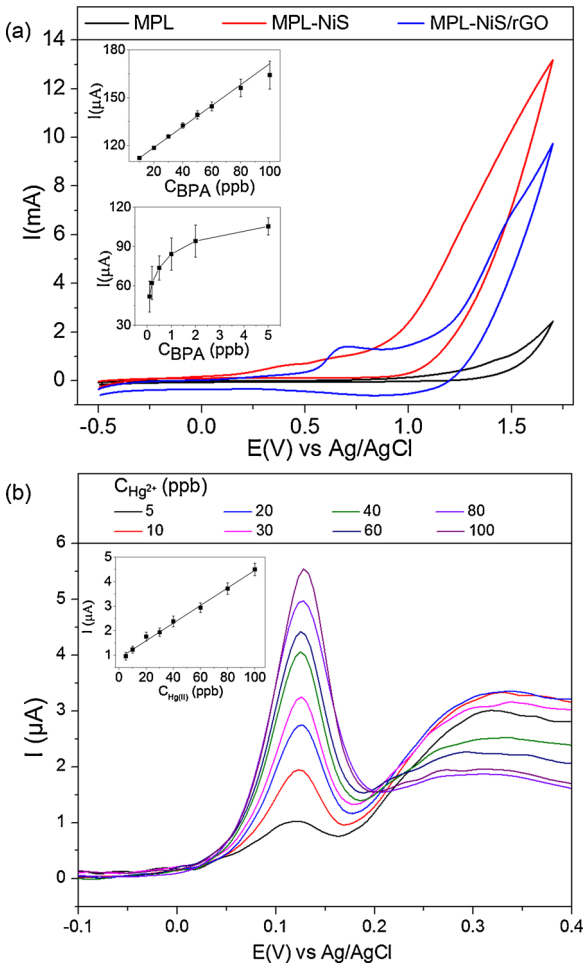

Fig. 3. (a) Cyclic voltammograms of a 0.1 ppm BPA sample separately measured using MPL, MPL-NiS, and MPL-NiS/rGO sensors. (Insets) Correlations between BPA concentration and peak intensity in the ranges of (top) 10–100 and (bottom) 0.1–5 ppb obtained by ASV. (b) ASV voltammograms acquired from Hg^{2+} samples with various concentrations using MPL-NiS/rGO. (b, inset) Correlation between peak intensity and Hg^{2+} concentration.

5 ppb. The intensity increased linearly with increasing concentration up to 60 ppb, and the relation was clearly nonlinear for concentrations of 80 ppb and greater, because of stripping process of BPA is non-equivalent. As shown in Fig. 3a, the irreversible oxidation peak of BPA was clearly observed and no reduction peak in reverse scan. The oxidation peak of BPA decreases gradually after successive scan because an insulating dimer layer formed during oxidation of BPA which diminishes the available active surface area. This passivation especially occurred during the analysis of high-concentration samples [21]. This explains the deviation of linearity for BPA concentrations greater than 80 ppb.

It is worthwhile to note that the measurements were conducted using three separately prepared MPL-NiS/rGO sensors. The average relative standard deviation (RSD) of the peak intensities in the triplicate measurements of each sample was 2.53%, so the sensor-to-sensor reproducibility was superior. The limit of detection (LOD= $3.3 \times \frac{SD}{b}$; *SD*: standard deviation of ordinate intercept, *b*: slope of regression line) calculated based on the response curve was 0.4 ppb (1.75 nM). The use of rGO made it possible to adsorb and accumulate BPA on the sensor surface, considerably contributing to the achieved sensitivity. The LOD achieved by the MPL-NiS/rGO was satisfactory, and comparable with the LODs noted in 9 previous reports regarding the electrochemical measurement of BPA using sensors incorporating graphene or carbon nanotubes (LODs ranging from 0.5 to 53 nM; refer to Table S1 in the

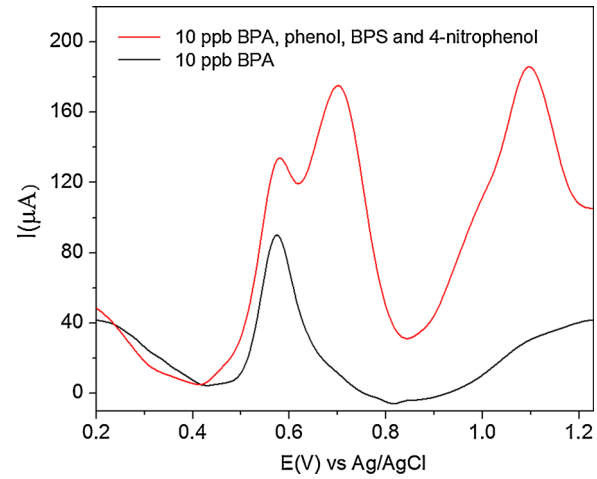

Fig. 4. ASV voltammograms acquired from a BPA sample contained with bisphenol S (BPS), 4-nitrophenol and phenol (concentrations of all components: 10 ppb) (red). Voltammogram obtained from a sample of 10 ppb BPA is also displayed (black).

Supplementary data). Also, the span of linear range (10–60 ppb, equivalent to 43 nM–0.26 μM) was moderate based on the comparison.

For the selectivity test, three phenol-derivative interferences of bisphenol S (BPS), 4-nitrophenol and phenol were selected. Then, a solution containing these interferences with BPA (concentrations of all components: 10 ppb) was measured. The resulting ASV voltammogram is shown in Fig. 4. The voltammogram obtained from a sample of 10 ppb BPA is also displayed for the comparison. In the measurement of the mixture solution, the BPA peak at 0.6 V is clearly apparent; while, the peaks around 0.7 and 1.1 V are also appeared. The peak at 0.7 V, close to the BPA peak, is originated from phenol and the peak at 1.1 V is contributed from both BPS and 4-nitrophenol. The locations of phenol, BPS and 4-nitrophenol peaks were confirmed by measuring each pure sample using the same sensor. Based on the observation, phenol could be an interfering component in the measurement of BPA; while, the interferences by BPS and 4-nitrophenol expect to be minimal. As also findable in a previous report [24–26], the oxidation peak of phenol was observed very near to 0.6 V in ASV measurement. By the way, based on a former report [27], phenol would not exist in extracted solutions from DVD and plastic-coated receipt paper. Therefore, no significant interference by phenol is expected in the measurements of real extracted samples.

The same three sensors were further used to measure BPA in extract solutions from real samples, namely a DVD and a piece of thermal receipt paper. A standard addition method was used to determine the concentrations of BPA in both samples, and HPLC was used to measure BPA concentrations of the same samples as reference measurements (also triplicate measurements). In the case of the DVD extract, the BPA concentrations determined using HPLC and the proposed sensor were 185.0 (RSD: 7.8%) and 172.0 ppb (RSD: 7.2%), respectively, corresponding to the method-to-method recovery of 93.0%. Similarly, the recovery was 91.3% for the receipt paper extract (HPLC: 126.0 ppm (RSD: 5.6%); proposed sensor: 115.0 ppm (RSD: 6.9%)). The recoveries in both cases were good, but below 100%. The deficits most probably originated from decreases in the electroactive surface area arising from adsorption of other substances existing in the extract solutions.

3.3. Measurement of Hg^{2+} samples using MPL-NiS/rGO sensor

To demonstrate the wider applicability of the sensor, Hg^{2+} samples in the concentration range from 5 to 100 ppb were also analyzed by means of ASV using the MPL-NiS/rGO sensor (Fig. 3b). The intensity of the peak at approximately 0.12 V steadily increased with higher Hg

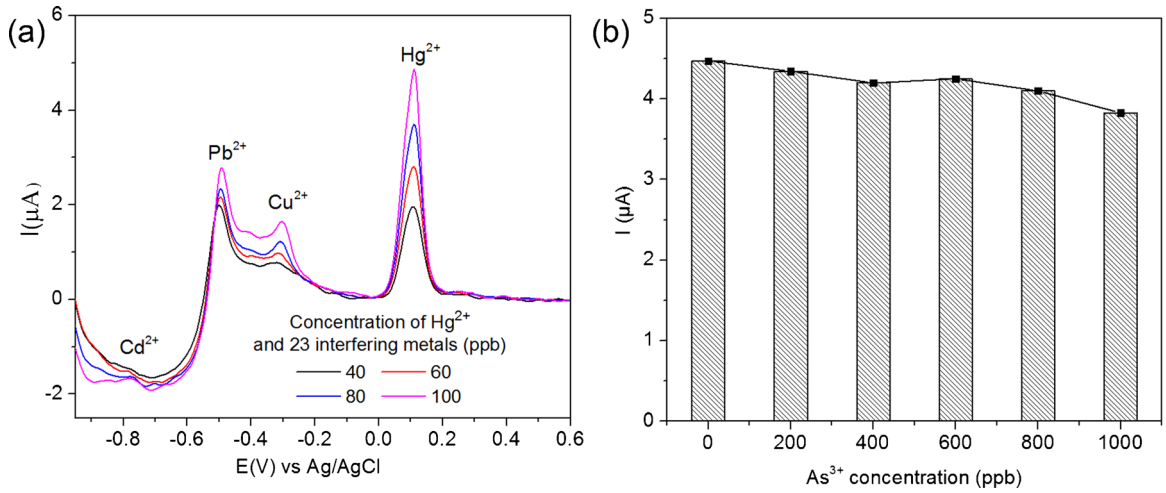

Fig. 5. (a) ASV voltammograms acquired from four different Hg^{2+} samples contained with 23 interfering ions (concentrations of all metal ions: 40, 60, 80 to 100 ppb). (b) Change of Hg^{2+} peak intensity under the presence of As^{3+} with varying concentrations.

concentration. The correlation between the intensity of this peak and the Hg^{2+} concentration was clearly linear, with the LOD of 0.16 ppb (0.8 nM; Fig. 3b inset). The measurements were accomplished using three separately prepared MPL-NiS/rGO sensors. Again, the average RSD of the peak intensities in the triplicate measurements of each sample was good, at 8.5%. Next, a 40 ppb Hg^{2+} sample were measured twenty-five times continuously using the proposed sensor. The peak intensity was steady during the first 10 measurements and decayed thereafter (Fig. S5), indicating the good stability and repeatability of measurements. As listed in Table S2, the analytical performance and LOD in Hg^{2+} analysis were comparable with those reported in previous publications regarding the use of graphene-based nanosensors to measure Hg^{2+}. The linear range (5–100 ppb, equivalent to 25 nM–0.5 μM) was relatively wider based on the comparison.

For the Hg^{2+} selectivity test, an element calibration standard solution containing Ag^{+}, Al^{3+}, B^{3+}, Ba^{2+}, Bi^{5+}, Ca^{2+}, Cd^{2+}, Co^{4+}, Cr^{6+}, Cu^{2+}, Fe^{3+}, Ga^{3+}, In^{3+}, K^{+}, Li^{+}, Mg^{2+}, Mn^{4+}, Na^{+}, Ni^{2+}, Pb^{4+}, Sr^{2+}, Tl^{3+} and Zn^{2+} ions (a total of 23 interfering ions) was initially used. Four different samples including all of the ions with Hg^{2+} (concentrations: 40, 60, 80 to 100 ppb) were prepared and measured. The resulting voltammograms are shown in Fig. 5a. The Cd^{2+}, Pb^{2+} and Cu^{2+} peaks are observable at −0.78, −0.5 and −0.3 V, respectively, as also observed in a previous publication [28]. These interfering peaks are away from the Hg^{2+} peak at 0.12 V. Therefore, the tested 23 metals ions would not seriously interfere the measurement of Hg^{2+}.

By the way, based on former reports [29,30], the As^{3+} peak appeared around 0.1 V, fairly near to the Hg^{2+} peak. Therefore, the potential interference by As^{3+} was further examined by measuring 100 ppb Hg^{2+} samples containing As^{3+} from 200 ppb to 1000 ppb. Fig. 5b shows the change of Hg^{2+} peak intensity under the presence of As^{3+} with varying concentrations. The intensity decreased by the interfering As^{3+} and up to approximately 14.4% when the As^{3+} concentration was 1000 ppb (10 times larger than the concentration of Hg^{2+}). Meanwhile, As^{3+} concentrations in real samples expect to be substantially low; therefore, the interference by As^{3+} in the measurement of Hg^{2+} would be marginal.

4. Conclusions

We demonstrated the analytical utility of MPL-NiS/rGO sensor that effectively incorporated NiS between MPL and rGO. The MPL-NiS/rGO structure is inexpensive and easy to make. A small and portable analytical device embed with the structure as a disposable electrochemical sensor is under development for on-site measurement of BPA. Also, sensors incorporating transition metal sulfides could be promising candidates for diverse electrochemical measurements, so we are searching for other transition metal sulfides that are suitable for various electrochemical analysis.

Acknowledgements

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.snb.2018.08.139.

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